

Tutorial Set 1

Q. The future lifetime r.v (T) follows a pareto distribution with pdf:

$$f(t) = \frac{\lambda m^\lambda}{t^{\lambda+1}}$$

Find the survival and hazard functions.

Solution.

$$S(t) = \int_t^\infty f(s) ds$$

$$= \int_t^\infty \lambda m^\lambda s^{-(\lambda+1)} ds$$

$$= -\frac{\lambda m^\lambda s^{-\lambda}}{\lambda} \Big|_t^\infty$$

$$= -m^\lambda s^{-\lambda} \Big|_t^\infty$$

$$= \left(\frac{m}{t}\right)^\lambda$$

$$h(t) = \frac{f(t)}{S(t)}$$

$$= \frac{\lambda m^\lambda t^{-(\lambda+1)}}{m^\lambda t^{-\lambda}}$$

$$= \lambda t^{-1}$$

$$= \frac{\lambda}{t}$$

2. The pdf of a future lifetime random variable (T) is

$$f(t) = \frac{x}{\theta^2} e^{-\frac{x^2}{2\theta^2}}$$

find the survival and hazard functions.

Solution.

$$S(t) = \int_t^{\infty} f(x) dx$$

$$= \int_t^{\infty} \frac{x}{\theta^2} e^{-\frac{x^2}{2\theta^2}} dx$$

$$= \frac{1}{\theta^2} \int_t^{\infty} x e^{-\frac{x^2}{2\theta^2}} dx$$

Using Substitution:

$$\text{Let } u = e^{-\frac{x^2}{2\theta^2}}$$

$$\frac{du}{dx} = -\frac{x}{\theta^2} e^{-\frac{x^2}{2\theta^2}}$$

$$\frac{du}{dx} = -\frac{x}{\theta^2} u$$

$$\Rightarrow dx = \frac{-du}{\frac{x}{\theta^2} u} = \frac{-\theta^2 du}{xu}$$

by substitution, we have

$$S(t) = \frac{1}{\theta^2} \int_t^{\infty} \cancel{x} u \cdot \frac{-\theta^2 du}{\cancel{x} u}$$

$$= - \int_t^{\infty} 1 du = -u \Big|_t^{\infty}$$

$$= -e^{-\frac{x^2}{2\theta^2}} \Big|_t^{\infty} = e^{-\frac{t^2}{2\theta^2}}$$

$$h(t) = \frac{f(t)}{g(t)}$$

$$= \frac{t}{t^2}$$

$$= \frac{1}{t}$$

$$= \frac{1}{t}$$

$$= \frac{1}{t}$$

Q3. Given the following cdfs, find the hazard function.

a). $F(t) = 1 - e^{(-\theta_0 t - \frac{1}{2}\theta_1 t^2)}$

Solution.

We know that:

$$F(t) + S(t) = 1$$

$$\Rightarrow S(t) = 1 - F(t)$$

$$\begin{aligned}\Rightarrow S(t) &= 1 - \left[1 - e^{(-\theta_0 t - \frac{1}{2}\theta_1 t^2)} \right] \\ &= e^{(-\theta_0 t - \frac{1}{2}\theta_1 t^2)}\end{aligned}$$

$$h(t) = \frac{f(t)}{S(t)}$$

$$f(t) = -\frac{d}{dt} S(t)$$

$$= -\frac{d}{dt} e^{(-\theta_0 t - \frac{1}{2}\theta_1 t^2)}$$

$$= -(-\theta_0 - \theta_1 t) e^{(-\theta_0 t - \frac{1}{2}\theta_1 t^2)}$$

$$= (\theta_0 + \theta_1 t) e^{(-\theta_0 t - \frac{1}{2}\theta_1 t^2)}$$

$$h(t) = \frac{(\theta_0 + \theta_1 t) e^{(-\theta_0 t - \frac{1}{2} \theta_1 t^2)}}{e^{(-\theta_0 t - \frac{1}{2} \theta_1 t^2)}} \\ = \theta_0 + \theta_1 t$$

b. $F(t) = 1 - e^{-(\frac{t}{\beta})^2}$

Solution

$$F(t) + S(t) = 1$$

$$\Rightarrow S(t) = 1 - F(t)$$

$$= 1 - [1 - e^{-(\frac{t}{\beta})^2}] \\ = e^{-(\frac{t}{\beta})^2}$$

$$f(t) = -\frac{d}{dt} S(t)$$

$$= -\frac{d}{dt} e^{-(\frac{t}{\beta})^2}$$

$$= -\left(-2\left(\frac{t}{\beta}\right)^{2-1}\left(\frac{1}{\beta}\right)\right) e^{-(\frac{t}{\beta})^2}$$

$$= \left(\frac{2}{\beta}\right)\left(\frac{t}{\beta}\right)^{2-1} e^{-(\frac{t}{\beta})^2}$$

$$= \frac{(\frac{2}{\beta})(\frac{t}{\beta})^2}{(\frac{t}{\beta})} e^{-(\frac{t}{\beta})^2} = \left(\frac{2}{t}\right)\left(\frac{t}{\beta}\right)^2 e^{-(\frac{t}{\beta})^2}$$

$$h(t) = \frac{f(t)}{g(t)}$$

$$= \left(\frac{2}{t} \right) \left(\frac{t^2}{\beta} \right) e^{-\left(\frac{t}{\beta} \right)^2}$$

$$= \frac{2}{t} \left(\frac{t^2}{\beta} \right)$$

Q4. The pmf of the future lifetime random variable (T) is

$$f(t) = q^t p, \text{ where } q = 1 - p$$

Find the survival and hazard functions.

Solution

$$f(t) = q^t p$$

$$\Rightarrow F(t) = P(T \leq t)$$

$$\Rightarrow S(t) = P(T > t)$$

$$= P(T \geq t+1)$$

$$= \sum_{S=t+1}^{\infty} q^S p$$

$$= p \left[q^{t+1} + q^{t+2} + q^{t+3} + \dots \right]$$

clearly it follows geometric with

$$r = \frac{q^{t+2}}{q^{t+1}} = q^{t+2-t-1} = q$$

$$a = q^{t+1}$$

$$S(t) = S_{\infty} = \frac{a}{1-r}$$

$$= \left[\frac{q^{t+1}}{1-q} \right] p$$

but $1-q = p$

$$= \frac{q^{t+1}}{p} \cdot p$$

$$= q^{t+1}$$

$$\begin{aligned}
 h(t) &= \frac{f(t)}{s(t)} \\
 &= \frac{q^t p}{q^{t+1}} \\
 &= q^{t-t-1} p \\
 &= p/q
 \end{aligned}$$

Q5. The pmf of the future lifetime random variable (T) is $f(t) = q^{t-1} p$, where $q=1-p$. Find the survival and hazard functions.

Solution.

$$\begin{aligned}
 f(t) &= q^{t-1} p \\
 \Rightarrow F(t) &= P(T \leq t) \\
 \Rightarrow S(t) &= P(T > t) \\
 &= P(T \geq t+1) \\
 &= \sum_{s=t+1}^{\infty} q^{s-1} p \\
 &= p [q^t + q^{t+1} + q^{t+2} + \dots] \dots
 \end{aligned}$$

clearly, this is geometric progression with

$$r = \frac{q^{t+1}}{q^t} = \frac{q^{t+2}}{q^{t+1}} = q$$

$$a = q^t$$

$$\begin{aligned}
 S(t) &= \sum_{\infty} = p \left[\frac{q}{1-q} \right] \\
 &= p \left[\frac{q^t}{1-q} \right] \quad \text{but } 1-q = p \\
 &= p \frac{q^t}{p} \\
 &= q^t
 \end{aligned}$$

$$\begin{aligned}
 h(t) &= \frac{f(t)}{S(t)} \\
 &= \frac{q^{t-1} p}{q^t} \\
 &= q^{t-1-t} p \\
 &= \frac{p}{q}
 \end{aligned}$$